

Matrix transformations in 3D

Key Points

For a matrix, M , to visualise transforms not on the standard list, use the transform on unit vectors

$$\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

Some transforms can be found in the formula booklet.

Reflection in planes $x = 0$, $y = 0$ and $z = 0$ change the sign of their coordinate respectively

Basic Skills

Q1. Compute the following matrix calculations

$$\text{a) } \begin{pmatrix} 6 & 3 & 5 \\ -1 & 2 & 3 \\ -1 & 1 & 2 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \\ -1 \end{pmatrix} \quad \text{b) } \begin{pmatrix} 2 & -1 & 0 \\ 1 & 1 & 3 \\ -4 & 3 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 5 \\ -1 & 0 & 4 \\ -2 & 1 & 0 \end{pmatrix}$$

Q2. Write a matrix for the following 2D transforms

- a) A stretch in y by a scale factor of 5
- b) A rotation anticlockwise by 150°
- c) A reflection in the line $y = \sqrt{3}x$

Q3. Explain what happens to the coordinate $A=(5,-2,3)$ if it is reflected in the plane $x = 0$

Competency Skills

Q4. Describe the following matrix transformations

$$\text{a) } \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad \text{b) } \begin{pmatrix} 1 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad \text{c) } \begin{pmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{pmatrix} \quad \text{d) } \begin{pmatrix} \cos \frac{\pi}{4} & 0 & \sin \frac{\pi}{4} \\ 0 & 1 & 0 \\ -\sin \frac{\pi}{4} & 0 & \cos \frac{\pi}{4} \end{pmatrix} \quad \text{e) } \begin{pmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} & 0 \\ \frac{1}{2} & \frac{\sqrt{3}}{2} & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Q5. Determine the matrix which represents a rotation of 180° about the x -axis

Q6. Why do we use planes to reflect in and not lines

Q7. If coordinate $A=(4,-3,3)$ is reflected in the plane $z = 2$ what is the new coordinate.

Q8. Explain why it is hard to write a reflection in the plane $x = 1$ as a matrix multiplication

Advanced Skills

Q9. A transformation performs the following steps in order:

1. Reflect coordinates in the plane $x = 0$ (single transform X)
2. Reflect coordinates in the plane $y = 0$ (single transform Y)
3. Reflect coordinates in the plane $z = 0$ (single transform Z)

- a) Suggest a matrix M which performs these transformations and use Mx to confirm it works
- b) A student suggests "We could multiply matrices X,Y,Z together to get M so $M = XYZ$ " confirm if this is true for this transformation
- c) What sort of transforms combined do you think this multiplying technique works for

Extension

Q10. Deduce a general method for transform A followed by transform B to form transform M